Integrare numerică

$f: I \to \mathbb{R}$ $\quad I = [a, b]$ → interval compact

f → continuă

F → primitivă → NU o cunosc

$f \to (x_i, f_i)$ cunoscută

↳ set de puncte

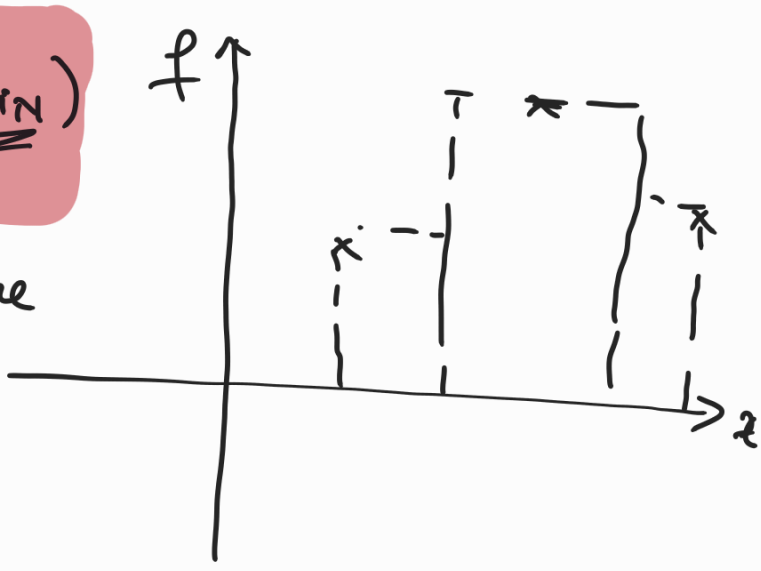
$$I[f] = \int_a^b f(x)\,dx = ?$$

↳ funcțională

problemă continuă → problemă discretizată

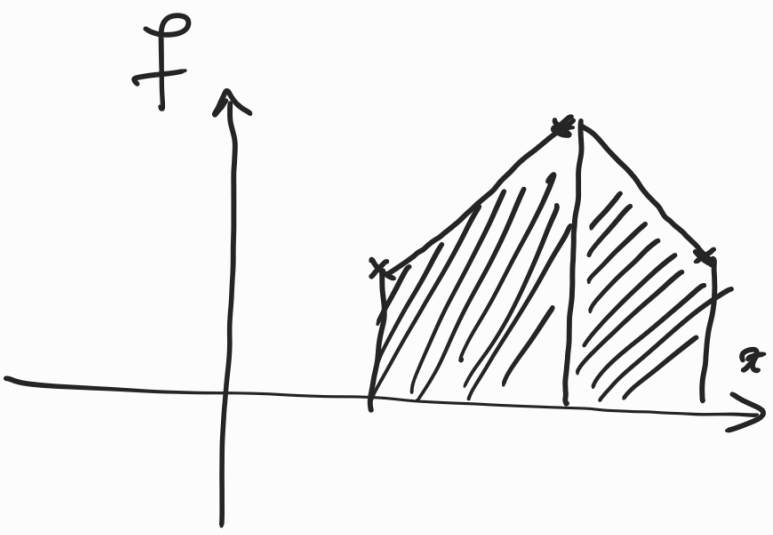
$$I_N[f] = \sum_{i=1}^{N} A_{iN} \cdot f(x_{iN})$$

↳ formulă de integrare



ex: $x_i = a + i \cdot h$

$h = \frac{b-a}{N}$, $i = \overline{0:N}$



Teoremă → Condiția necesară și suficientă pentru ca $I_N[f]$ să converge $I[f]$.

slab !!

se exprimă prin:

a) $\exists M > 0$ a.î. $\sum_{i=1}^{N} |A_{iN}| < M$ pt. $N = 1, 2, 3, \ldots$

↳ să nu existe creșteri infinite

b) $\lim_{N \to +\infty} I[x^k] = \int_a^b x^k dx$

↳ polinom de grad k

↓ formula este exactă pt. polinoame

I. Metode Newton-Cotes de integrare

$$\int_a^b f(x) \cdot w(x)\, dx = \sum_{k=1}^{N} A_{iN} \cdot f(x_{iN}) + R_N$$

↳ funcție pondere

↳ eroare/rest

↙ depinde de N

Setul de puncte = set de puncte echidistante

$$x_{iN} = a + ih, \quad h = \frac{b-a}{N}, \quad i = 0:N$$

? $A_{iN} \rightarrow f(x) \cong P_{N-1}(x) = \sum_{i=1}^{N} f(x_{iN}) \cdot l_i(x)$

$$\int_a^b P_{N-1}(x)\,\omega(x)\,dx = \sum_{i=1}^{N} A_{iN} f(x_{iN})$$

$$\int_a^b \sum_{i=1}^{N} f(x_{iN}) \cdot l_i(x) \cdot \omega(x)\,dx =$$

$$\sum_{i=1}^{N} f(x_{iN}) \cdot \int_a^b l_i(x) \cdot \omega(x)\,dx$$

$$\Rightarrow A_{iN} = \int_a^b l_i(x) \cdot \omega(x)\,dx \quad \rightarrow \text{Formulă Newton Cotes}$$

$$l_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{N} \frac{x - x_j}{x_i - x_j}$$

$\int$/

$$f(x) = P_{N-1}(x) + E_{N-1}(x)$$

formula numerica: $I_N[f]$

$$\int_a^b f(x)\cdot\omega(x)\,dx = \int_a^b P_{N-1}(x)\,\omega(x)\,dx + \int_a^b E_{N-1}(x)\cdot\omega(x)\,dx$$

$f(x)\cdot\omega(x)\,dx$: $I[f]$

$\int_a^b E_{N-1}(x)\cdot\omega(x)\,dx$: R_N (rest)

↓ eroare de la interpolarea Newton

$$R_N = \frac{f^{(N)}(\xi)}{(N+1)!} \int_a^b (x-x_0)\cdot\ldots\cdot(x-x_n)\cdot\omega(x)\,dx$$

↓ formula erorii

(1) $N=1 \Rightarrow$

Formula trapezelor !

$$I_N[f] = \int_a^b f(x)\,dx = \frac{h}{2}(f(a)+f(b)) - \frac{h^3 f''(\xi)}{12}$$

$$h = \frac{b-a}{N} = b-a$$

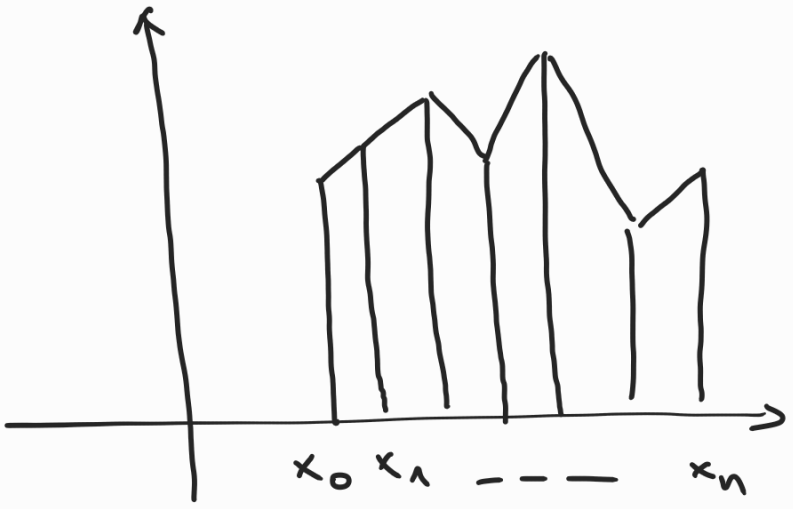
(2.) $N=2$

Formula Simpson !

$$I_N[f] = \int_a^b f(x)\,dx = \frac{h}{3}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right] - \frac{h^5 f'''(\xi)}{90}$$

Formula compusă

$$\int_{a=x_0}^{b=x_n} f(x)\,dx = \int_{x_0}^{x_1} + \int_{x_1}^{x_2} + \ldots + \int_{x_{n-1}}^{x_n}$$



Se aplică pe fiecare interval formula simplă a trapezelor / Simpsons.

$$I_N[f] = \frac{h}{2}\left[f(a) + f(b) + 2\sum_{i=1}^{N-1} f(a+ih)\right], \quad h = \frac{b-a}{N}$$

↓ trapeze

Formula compusă Simpson

$$I_N[f] = \frac{h}{3}\left[f(a) + f(b) + 4\sum_{i=1}^{N} f(x_{2i-1}) + 2\sum_{i=1}^{N-1} f(x_{2i})\right] \qquad h = \frac{b-a}{2N}$$

↳ ++ puncte intermediare

Eroare

→ trapeze $-\frac{h^3 f''(\xi)}{12} \cdot N = \frac{(b-a)^3 \cdot f(\xi)}{12 N^{\cancel{3}2}} \cdot \cancel{N} \sim \frac{1}{N^2}$

→ Simpson $-\frac{h^5 f'''(\xi)}{90} \cdot N = \frac{(b-a)^5 f'''(\xi)}{90 N^{\cancel{5}4}} \cdot \cancel{N} \sim \frac{1}{N^4}$

Cuadraturi gaussiene → punctele nu sunt echidistante

$$I[f] = \int_a^b f(x)\cdot\omega(x)\,dx$$

$$I_N[f] = \sum_{i=1}^{N} A_{iN}\cdot f(x_{iN}) + \text{(Teorema)}$$

↳ $2N$ necunoscute

$$k=0 \Rightarrow \sum_{i=1}^{N} A_{iN} = \int_a^b \omega(x)\,dx$$

$$k=1 \Rightarrow \sum_{i=1}^{N} A_{iN}\cdot x_{iN} = \int_a^b x^k\cdot\omega(x)\,dx$$

$$\vdots$$

$$\sum_{i=1}^{N} A_{iN}\cdot x_{iN}^k = \int_a^b x^k\omega(x)\,dx$$

$k = \overline{0:2N-1}$ $\qquad P = 1, x, x^2, \dots, x^{2n-1}$

Formăm $\Pi_N(x) = (x-x_{1N})(x-x_{2N})\cdot\ldots\cdot(x-x_{NN})$

$$P_{N+q-1}(x) = \Pi_N(x)\cdot Q_{q-1}(x) + R_{N-1}(x)$$

$N+q-1 \le 2N$

↳ teorema împărțirii cu rest

$$\int_a^b P_{N+q-1}(x)\cdot\omega(x)\,dx = \sum_{i=1}^{N} A_{iN}\cdot P_{N+q-1}(x_{iN})$$

$$P_{N+q-1}(x_{iN}) = R_{N-1}(x_{iN})$$

$$\Rightarrow \int_a^b P_{N+q-1}(x)\,\omega(x)\,dx = \sum_{i=1}^{N} A_{iN}\cdot R_{N-1}(x_{iN})$$

$$\Leftrightarrow \int_a^b \pi_N(x)\cdot Q_{q-1}(x)\cdot\omega(x)\,dx + \int_a^b R_{N-1}(x)\,\omega(x)\,dx = \sum_{i=1}^{N} A_{iN}\cdot R_{N-1}(x_{iN})$$

$$\underbrace{\sum_{i=1}^{N} A_{iN}\,R_{N-1}(x_{iN})}$$

$$\Rightarrow \boxed{\int_a^b \pi_N(x)\cdot Q_{q-1}(x)\cdot\omega(x)\,dx = 0}$$

pentru toate valorile $q=\overline{0,N}$

↓ cele două polinoame sunt ortogonale

$\Rightarrow$ π_N este ortogonal cu orice polinom de grad mai mic !

a	b	$\omega(x)$	Polinom
-1	1	$\frac{1}{\sqrt{1-x^2}}$	Cebîşev
-1	1	1	Legendre
0	∞	e^{-x}	Laguave
$-\infty$	$+\infty$	e^{-x^2}	Hermite

$$\begin{cases} \int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}}\,dx = \frac{\pi}{N}\sum_{i=1}^{N} f(x_{iN}) \\ x_{iN} = \cos\left(\frac{2i-1}{2N}\pi\right) \end{cases}$$

$N=3$

$$\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}}\,dx = \frac{\pi}{3}\left(f\left(\frac{\sqrt{3}}{2}\right)+f(0)+f\left(-\frac{\sqrt{3}}{2}\right)\right)$$

$i=\overline{1,3}$ $\quad x_{1N} = \cos\left(\frac{1}{6}\pi\right) = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$

$$x_{2N} = \cos\left(\frac{3\pi}{6}\right) = \cos\frac{\pi}{2} = 0$$

$$x_{3N} = \cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$$

Formule de cuadraturi gaussiene

① Polinoame Cebâșev

$$a = -1$$
$$b = 1$$
$$\omega(x) = \frac{1}{\sqrt{1-x^2}}$$

$$\begin{cases} T_0(x) = 1 \\ T_1(x) = x \\ T_{n+1}(x) - 2x\,T_n(x) + T_{n-1}(x) = 0 \end{cases}$$

$$\int_{-1}^{1} f(x)\cdot \underbrace{\omega(x)}_{\frac{1}{\sqrt{1-x^2}}} dx = \underbrace{\frac{\pi}{N+1}}_{A_{iN}} \sum_{i=1}^{N} f(x_{iN})$$

↳ rădăcinile polinomului lui T_{N+1}

x_{iN} → rădăcinile polinom. Cebâșev =

$$= \cos\left(\frac{2i+1}{2N+2}\pi\right)$$

$$A_{iN} = \frac{\pi}{N+1}, \quad i = 0:N$$

2. Polinoame Legendre

$$\begin{cases} a=-1 \\ b=1 \end{cases} \qquad \omega(x)=1$$

$$\begin{cases} L_0(x)=1 \\ L_1(x)=x \end{cases}$$

$$(n+1)L_{n+1}(x)-(2n+1)xL_n(x)+nL_{n-1}(x)=0$$

$$\int_{-1}^{1} f(x)\,dx = \sum_{i=0}^{N} A_{iN}\cdot f(x_{iN})$$

x_{iN} = rădăcinile polinomului Legendre L_N

$$A_{iN}=\frac{2(1-x_{iN}^2)}{n^2 L_{n+1}^2(x_{iN})}$$

3. Polinoame Laguere

$a=0$

$b=\infty$

$\omega=e^{-x}$

$$\begin{cases} G_0(x)=1 \\ G_1(x)=1-x \\ G_{n+1}(x)-(2n+1-x)G_n(x)+ \\ n^2 G_{n-1}(x)=0 \end{cases}$$

$$\int_0^{\infty} f(x)e^{-x}\,dx = \sum_{i=0}^{N} A_{iN}\cdot f(x_{iN})$$

x_{iN} → rădăcinile polinomului G_n

$$A_{iN} = \frac{x_{iN}}{(n+1)^2 \, G_{n+1}^2(x_{iN})}$$

(4.) Polinoame Hermite

$a = -\infty$

$b = \infty$

$\omega(x) = e^{\frac{-x^2}{2}}$

$$\begin{cases} H_0(x) = 1 \\ H_1(x) = x \\ H_{n+1}(x) - x H_n(x) + n H_{n-1}(x) = 0 \end{cases}$$

$$\int_{-\infty}^{\infty} f(x) \cdot e^{-\frac{x^2}{2}} dx = \sum_{i=0}^{N} A_{iN} \cdot f(x_{iN})$$

x_{iN} rădăcinile lui H_N

$$A_{iN} = \frac{2^{n+1} \cdot n! \sqrt{\pi}}{H_{n+1}^2(x_{iN})}$$

Exemplu

$$\int_0^1 f(x)\,dx = a_0 f(0) + a_1 f(1) + b_0 f(x_0) + b_1 f(x_1)$$

$\Downarrow$ 6 necunoscute



Grad maxim de valabilitate

↳ $2 \cdot 4 - 1 = (7)$

$$\Pi_4(x) = (x-0)(x-1)(x-x_0)(x-x_1) =$$

$\Downarrow$ GRAD DE VALABILITATE = (7)

$$= (x^2 - x)(x^2 - sx + p)$$

$$\begin{cases} \int_0^1 \Pi_4(x) \cdot x^0\,dx = 0 \\ \int_0^1 \Pi_4(x) \cdot x^1\,dx = 0 \end{cases} \iff$$

$$\begin{cases} \int_0^1 (x^2 - x)(x^2 - sx + p)\,dx = 0 \\ \int_0^1 (x^3 - x^2)(x^2 - sx + p)\,dx = 0 \end{cases}$$

calcul integrală $\Rightarrow$

$$\begin{cases} \frac{1}{5} - \frac{S}{4} + \frac{P}{3} - \frac{1}{4} + \frac{S}{3} - \frac{P}{2} = 0 \\ \frac{1}{6} - \frac{S}{5} + \frac{P}{4} - \frac{1}{5} + \frac{S}{4} - \frac{P}{3} = 0 \end{cases} \Rightarrow$$

$$\begin{cases} \frac{S}{12} - \frac{P}{6} = \frac{1}{20} \\ \frac{S}{20} - \frac{P}{12} = \frac{1}{30} \end{cases} \Rightarrow$$

$$\begin{cases} \frac{S}{12} - \frac{P}{6} = \frac{1}{20} \\ \frac{S}{10} - \frac{P}{5} = \frac{1}{15} \end{cases} \Rightarrow \quad \frac{S}{10} - \frac{S}{12} = \frac{1}{15} - \frac{1}{20}$$

$$\frac{S}{60} = \frac{1}{60} \Rightarrow \boxed{S = 1}$$

$$\frac{1}{12} - \frac{P}{6} = \frac{1}{20} \Rightarrow \frac{P}{6} = \frac{1}{12} - \frac{1}{20} \Rightarrow$$

$$\boxed{P = \frac{1}{5}}$$

$$\Rightarrow \quad x^2 - x + \frac{1}{5} = 0$$

$$\Delta = 1 - \frac{4}{5} = \frac{1}{5}$$

$$x_{1,2} = \frac{1 \pm \frac{\sqrt{5}}{5}}{2} = \frac{5 \pm \sqrt{5}}{10}$$

$$x_0 = \frac{5 - \sqrt{5}}{10}$$

$$x_1 = \frac{5 + \sqrt{5}}{10}$$

$$\int_0^1 f(x)\,dx = a_0 f(0) + a_1 f(1) + b_0 f\left(\frac{5-\sqrt{5}}{10}\right)$$

$$+ b_1 \cdot f\left(\frac{5+\sqrt{5}}{10}\right)$$

Formula este valabilă pt. polinoame de grad $1, 2, \dots, 7$.

$f = 1 \quad a_0 + a_1 + b_0 + b_1 = \int_0^1 dx = 1$

$f = x \quad a_0 \cdot 0 + a_1 + b_0 \cdot \frac{5-\sqrt{5}}{10} + b_1 \cdot \frac{5+\sqrt{5}}{10} =$

$= \int_0^1 x\,dx = \frac{1}{2}$ (*)

$f = x^2 \quad a_1 + b_0 \cdot \left(\frac{5-\sqrt{5}}{10}\right)^2 + b_1 \cdot \left(\frac{5+\sqrt{5}}{10}\right)^2 =$

$= \int_0^1 x^2 dx = \frac{2}{3}$

(*) $\begin{cases} a_1 + \frac{b_0}{2} + \frac{b_1}{2} = \frac{1}{2} \\ b_0 = b_1 \end{cases}$

(**) $b_0 = b_1$

$$a_1 + \frac{30 - 10\sqrt{5}}{100} + b_1 \cdot \frac{30 + 10\sqrt{5}}{100} = \frac{1}{3}$$

$$\begin{cases} a_1 + \frac{3}{10} b_0 + \frac{3}{10} b_1 = \frac{1}{3} \\ b_0 = b_1 \end{cases}$$

$$\begin{cases} a_0 + a_1 + b_0 + b_1 = 1 \\ b_0 = b_1 \\ a_1 + \frac{b_0}{2} + \frac{b_1}{2} = \frac{1}{2} \\ a_1 + \frac{3}{10} b_0 + \frac{3}{10} b_1 = \frac{1}{3} \end{cases} \Rightarrow \begin{cases} a_0 + a_1 + 2b_0 = 1 \\ a_1 + b_0 = \frac{1}{2} \\ a_1 + \frac{3}{5} b_0 = \frac{1}{3} \end{cases}$$

$$\Rightarrow \frac{2}{5} b_0 = \frac{1}{6} \Rightarrow \begin{cases} b_0 = \frac{5}{12} \\ b_1 = \frac{5}{12} \end{cases}$$

$$a_1 = \frac{1}{2} - b_0 = \frac{1}{12} \Rightarrow \boxed{a_1 = \frac{1}{12}}$$

$$\boxed{a_0 = \frac{1}{12}}$$

$$\int_0^1 f(x)\,dx = \frac{1}{12}\left[f(0) + f(1) + 5f\left(\frac{5-\sqrt{5}}{10}\right) + 5f\left(\frac{5+\sqrt{5}}{10}\right)\right]$$





Exemple





1) $\int_0^1 f(x)\,dx = K\left[f(x_1) + f(x_2) + f(x_3)\right]$

2) $\int_{-1}^{1} f(x)\,dx = a_0 f(-1) + b_0 f(1) + c_0 f(x_0)$

3) $\int_{-1}^{1} f(x)\,dx = a_0 f(-1) + b_0 f(0) + c_0 f(1)$

↳ comparați cu trapeze

4) $\int_{0}^{1} f(x)\,dx = a_0 f(x_0) + b_0 f(x_1) + a_1 f'(x_0) +$

$b_1 f'(x_1)$